The EV Transition in India: A Value Chain and Policy Impact Analysis

What is the EV industry?

The **Electric Vehicle (EV) industry** encompasses the research, development, manufacturing, infrastructure, and sales of vehicles that are fully or partially powered by electric energy, aiming to replace traditional fossil-fuel-based transportation.
It is a rapidly evolving sector driven by technological advancements and global environmental goals to reduce carbon emissions and dependence on petroleum.

1830s – 1890s

Early Competition. The first electric carriages emerged in the 1830s, predating functional gasoline cars. By 1900, EVs were at their "heyday," making up about **one-third of all vehicles** on US roads due to their silent operation compared to noisy steam and early ICE models.

1910s – 1920s

The Rise of ICE. Several factors led to the near-disappearance of EVs: the Ford Model T (1908) made gasoline cars affordable, and the invention of the electric starter (1912) eliminated the difficult hand crank. Improved roads and cheap, plentiful oil further favored the longer range of ICE vehicles.

1970s – 1990s

Renewed Interest. Global oil crises in the 1970s sparked experimental EV prototypes. In the 1990s, the Toyota Prius (1997) launched as the first mass-produced hybrid, and GM released the EV1 (1996), signaling a shift back toward electrification.

2000s – 2020

The Modern Era. The 2008 Tesla Roadster proved EVs could be high-performance, and the subsequent Nissan LEAF (2010) brought all-electric driving to the mass market.

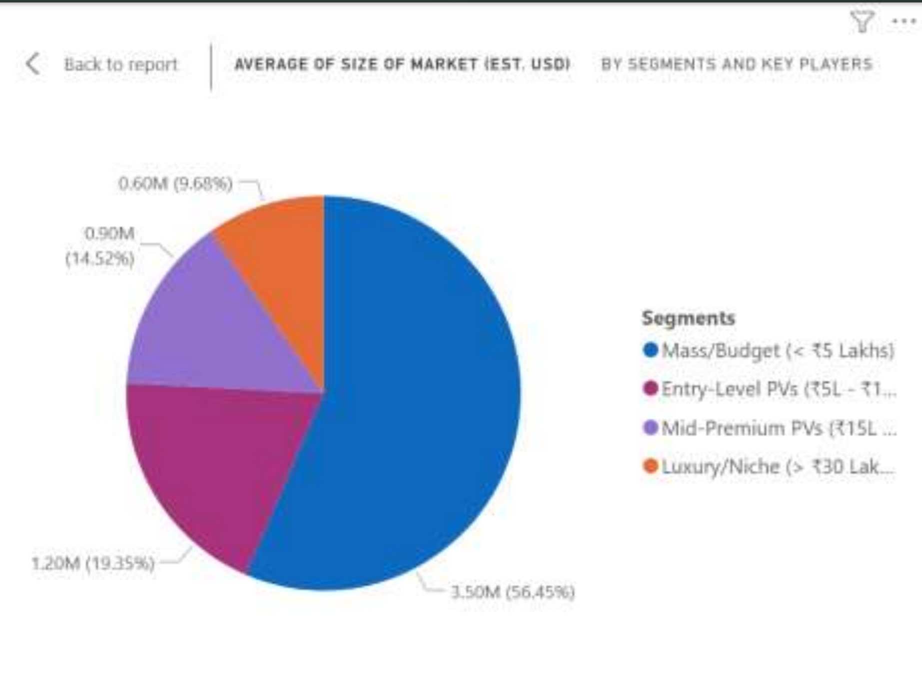
2021 – 2025

The Acceleration Phase. In 2025, global sales continue to surge, with ICE production in some regions (like North America) projected to decline by more than half by 2029. Governments and manufacturers have set deadlines, such as California's 100% zero-emission mandate for 2035

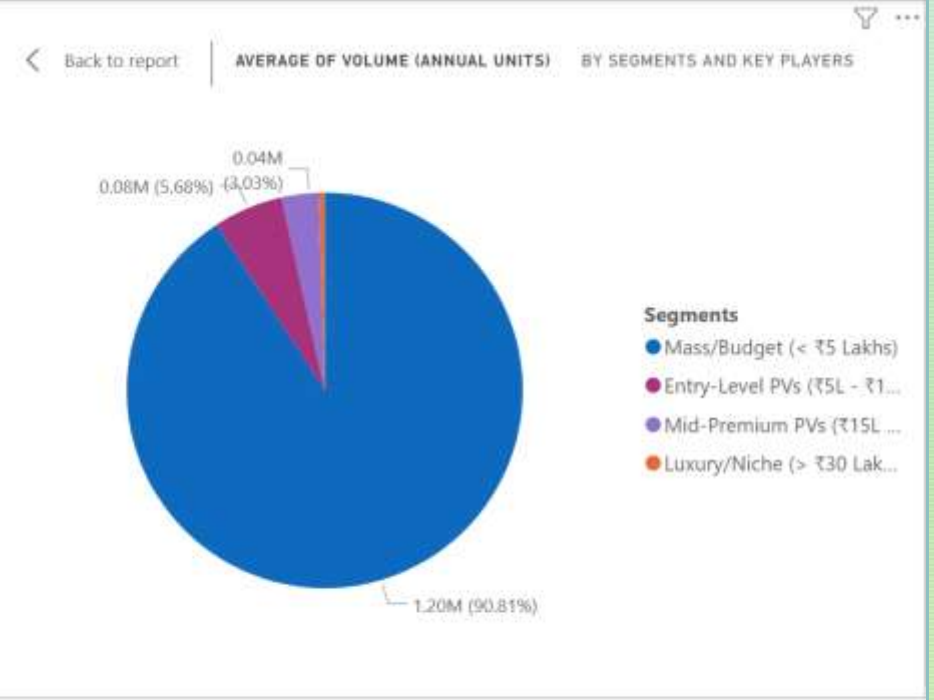
Historical timeline of transition from ICE to EV

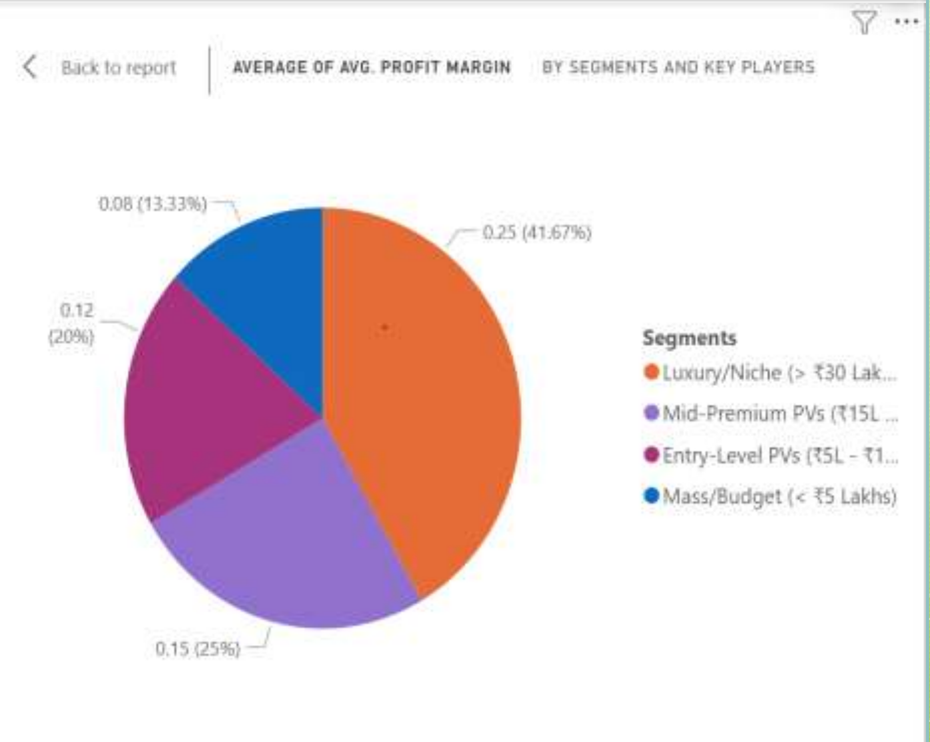
Market Analysis

- *The **Volume** is highest in the Budget segment (2W/3W), but the **Profit Margin** per unit is highest in the Luxury segment. This is a classic "High Volume, Low Margin" vs "Low Volume, High Margin" business strategy comparison.*

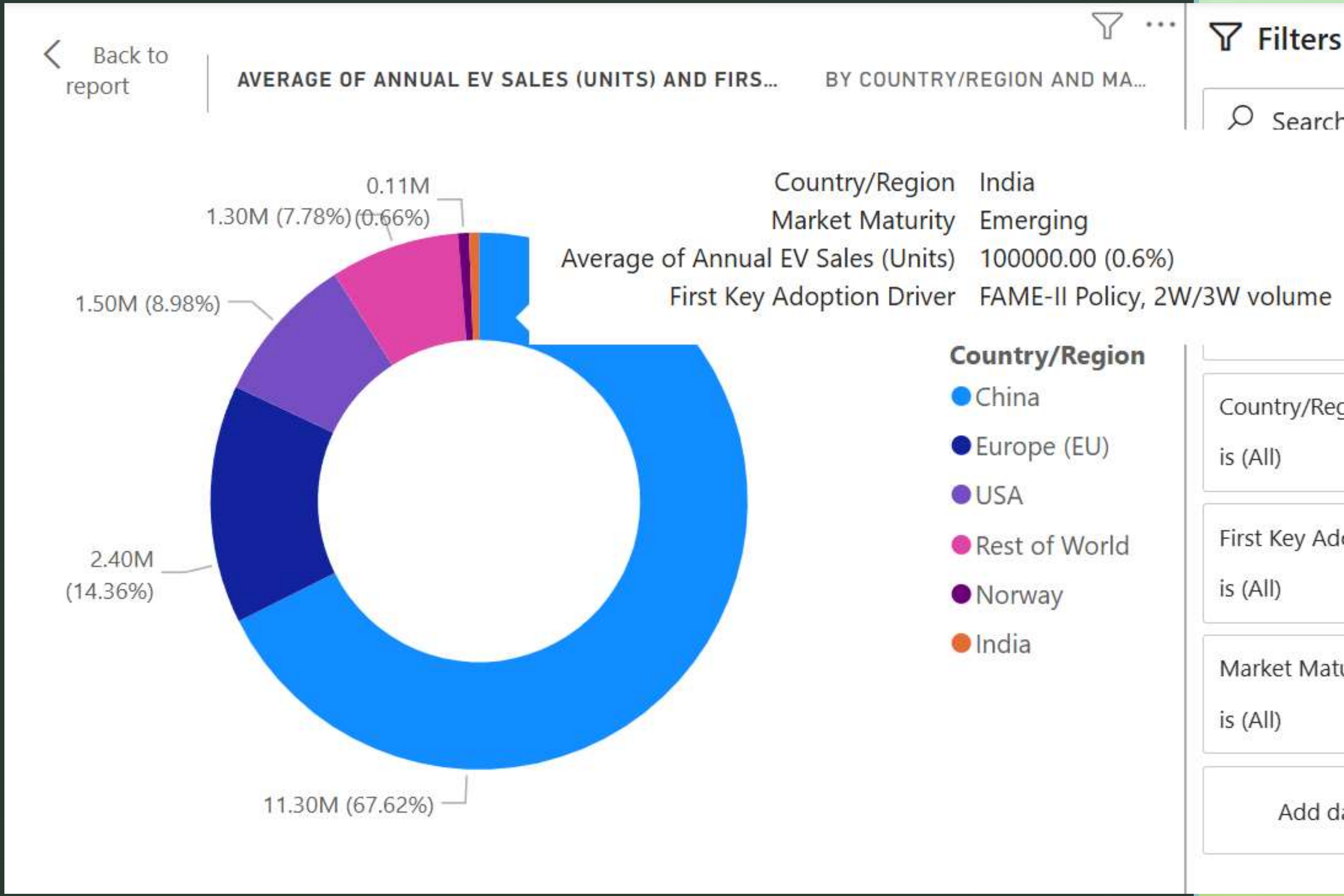


- *While 2-wheelers dominate the volume, the profitability and long-term infrastructure demand will be driven by the 4-wheeler Passenger Vehicle (PV) segments.*





- *In 2024, nearly 1* ***out of every 2 cars*** *sold in China was electric.*
- *While India's car market share is low (2%), the* ***2-Wheeler (E-2W)*** *and* ***3-Wheeler (E-3W)*** *market shares are significantly higher (approaching 5-7%). India is "skipping" the car-first transition and leading in* ***Micro-mobility****.*
- *The USA has a smaller volume than China but a higher "Average Selling Price" (ASP) because most EVs sold there are premium SUVs/luxury cars.*
- *The Indian EV market is projected to reach $100B by 2030").*



Current size of the EV market globally

PESTEL Analysis

- **Political: The "Policy Beta" Effect**

The Power of Policy: Academic research shows that **Political Factors** (measured with a coefficient of 0.82) are the single strongest predictor of EV adoption—outperforming even direct economic subsidies.

- **Economic: The "Price Parity" Myth**

The Tipping Point: While many think EVs are "too expensive," the **Total Cost of Ownership (TCO)** for an electric 3-wheeler in India is already **30-40% lower** than its diesel counterpart. This is why the 3-wheeler segment is the first to reach 50%+ electrification in some Indian cities.

- **Social: Range Anxiety vs. Charging Reality**

The "Psychological" Range: Interestingly, while consumers demand a **500 km+ range**, data shows that **95% of daily commutes** in Indian metros are under **50 km**. The gap is a social perception issue, not a technical one.

- **Technological: The Solid-State Revolution**

Beyond Lithium-Ion: Solid-state batteries (SSBs) are expected to enter mass production by **2027-2030**. They are "game-changers" because they use solid electrolytes, making them **non-flammable** and capable of charging to 80% in under **10 minutes**.

Category	Key Factors & Strategic Impact	Important Numerical Values (2025)
Political	Transition from FAME-II to **PM E-DRIVE**. Shift from simple subsidies to mandates for public transport & charging infrastructure.	**₹10,900 Cr:** PM E-DRIVE budget. **114% :** Increase in budget over previous years. **100% :** Subsidy for charging in govt. buildings.
Economic	Battery costs approaching ICE price parity. High growth in micro-mobility (2W/3W) segments due to lower TCO (Total Cost of Ownership).	**$108/kWh:** Global avg. battery pack price. **$80/kWh:** Target for price parity (est. 2026). **15% +:** Projected CAGR for Indian EV market.
Social	Increasing urban acceptance. Adoption driven by "green" branding and significantly lower running costs per km.	**20 Lakh+:** Annual EV sales in India (2025). **50% :** Sales share of E-2Wheelers in total EVs. **300% :** Surge in demand for service infrastructure.
Technological	Evolution of battery chemistries (LFP dominance). Integration of **AI & IoT** in scooters (Smart BMS). Focus on Fast Charging.	**40,000:** Charging points target for Tata by 2027. **72,300:** Public chargers targeted by PM E-DRIVE. **50% :** Higher energy density in prototype solid-state cells.
Environmental	Stricter LCA (Life Cycle Assessment) scrutiny. Regulatory push for **Circular Economy** (Battery recycling and reuse).	**73% :** Reduction in GHG emissions vs ICE. **24,000 km:** Avg. "Carbon Parity" distance for EVs. **Net-Zero 2070:** India's long-term climate goal.
Legal	Stringent battery safety norms (**AIS 156 Phase 2**). CAFE (Corporate Average Fuel Efficiency) targets for manufacturers.	**AIS 156:** Mandatory thermal/pressure tests. **50% :** Min. Domestic Value Addition (DVA) for PLI. **IPX7:** Mandatory water-ingress protection rating.

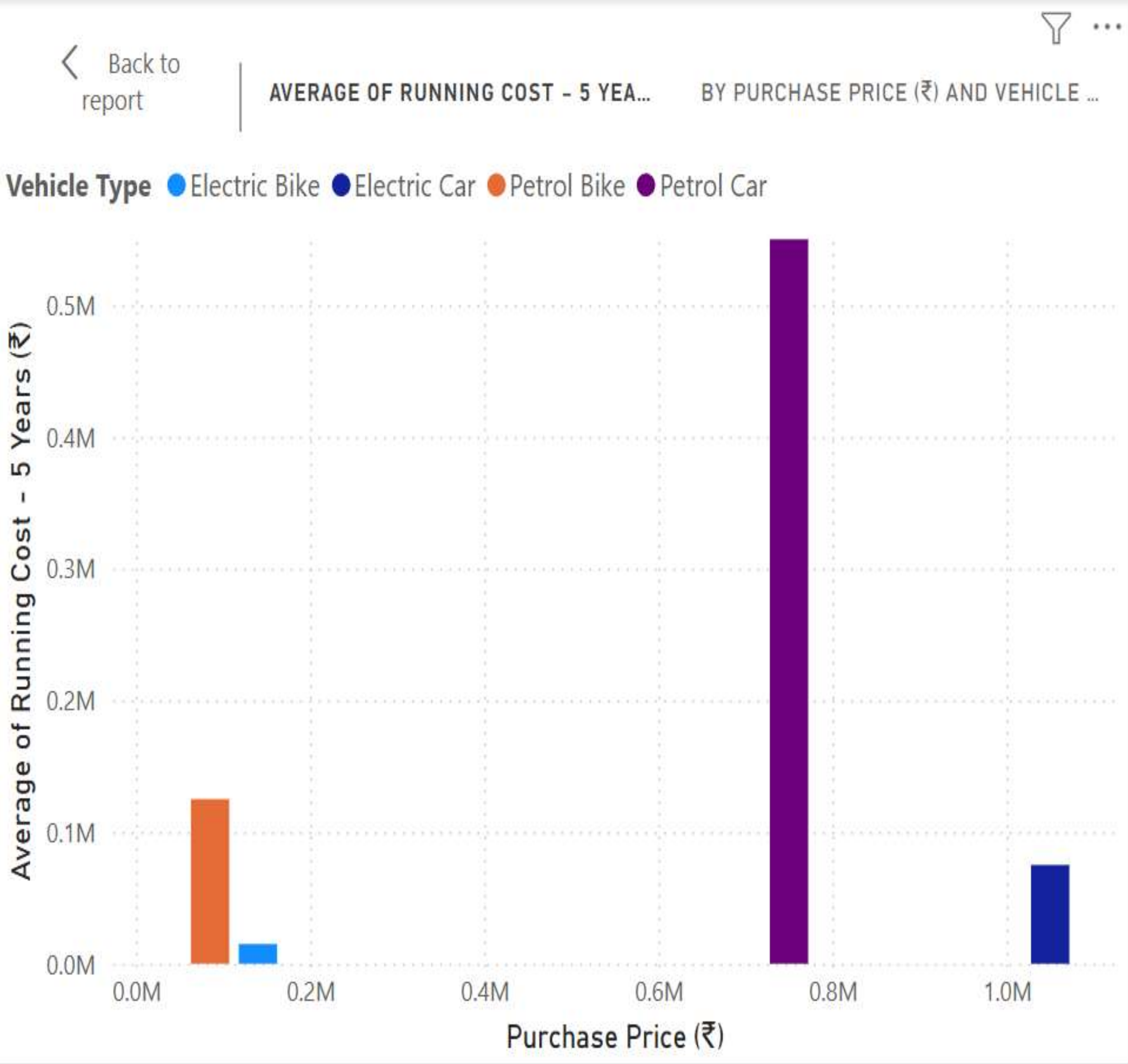
PESTEL analysis of current EV industry

- **Environmental: The "Carbon Debt"**

Manufacturing Deficit: Producing an EV battery is so energy-intensive that a new EV starts its life with a **"Carbon Debt."** It typically takes **15,000 to 24,000 km** of driving (about 1–2 years) before the EV becomes "cleaner" than a petrol car. **Water Intensity:** Producing just 1 tonne of Lithium requires nearly **2 million litres of water**. This is creating environmental "hotspots" in the Lithium Triangle (Chile, Argentina, Bolivia).

- **Legal: The "Safety Pivot"**

AIS 156 Standards: After several fire incidents in 2022-23, India's **AIS 156 (Phase 2)** rules became some of the strictest in the world. Batteries must now survive being dropped, crushed, and heated to extreme temperatures, which has forced a "survival of the fittest" among Indian EV startups.



Total Cost of Ownership comparison of ICE vs EV

Value Chain Analysis

*A **Value Chain Analysis** breaks down the EV industry into stages that add value to the final product. It is categorized into **Upstream**, **Midstream**, and **Downstream** activities.*

1. Upstream: Raw Materials & Sourcing

This stage involves the "mining and refining" of the earth's minerals. It is the most geographically sensitive part of the chain.

- **Mining:** Extraction of critical minerals like **Lithium** (from the "Lithium Triangle" in South America), **Cobalt** (mostly from DR Congo), **Nickel**, **Manganese**, and **Graphite**.
- **Rare Earth Elements (REEs):** Neodymium and Dysprosium are mined for high-efficiency permanent magnet motors.
- **Refining & Processing:** Raw ores are chemically processed into high-purity battery-grade chemicals (e.g., Lithium Hydroxide). China currently controls ~60-80% of global refining capacity.

2. Midstream: Component Manufacturing

This is where the refined materials are transformed into the "guts" of the EV.

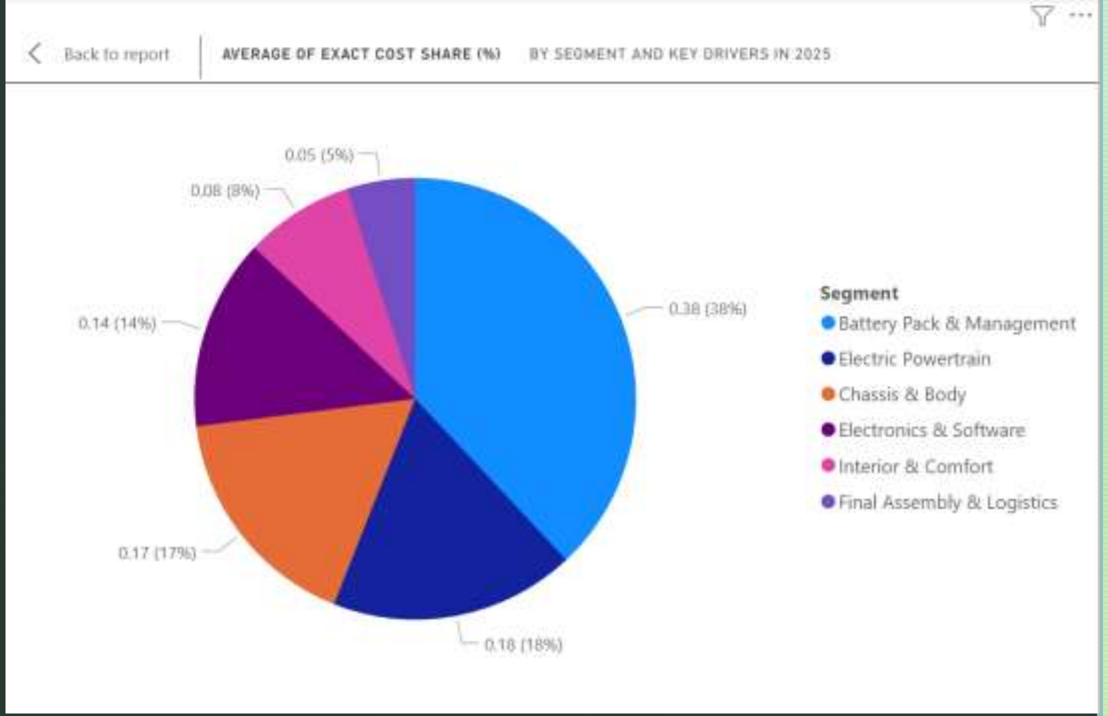
- **Battery Cell Manufacturing:** The most complex step. Cells are made of an **Anode, Cathode, Electrolyte, and Separator**.
- **Module & Pack Assembly:** Individual cells are grouped into modules, which are then integrated with a **Battery Management System (BMS)** and thermal cooling to form the final battery pack.
- **Power Electronics:** Production of Inverters (DC to AC), DC-DC converters, and Onboard Chargers.
- **Electric Motor Production:** Manufacturing of traction motors (mostly Permanent Magnet or Induction motors).

3. Downstream: Vehicle Assembly & Distribution

This is where traditional automotive expertise meets new-age tech.

- **OEM Assembly:** Manufacturers like **Tata Motors, Tesla, or BYD** integrate the chassis, body, and EV-specific powertrain into the final vehicle.
- **Sales & Marketing:** Utilizing dealerships or "Direct-to-Consumer" models (like Ola Electric) to reach the buyer.
- **Charging Infrastructure:** Deployment of home chargers and public fast-charging networks (a critical value-add for the ecosystem).

> *The EV value chain is unique because **30-40% of the vehicle's value** is concentrated in a single component: the battery.*



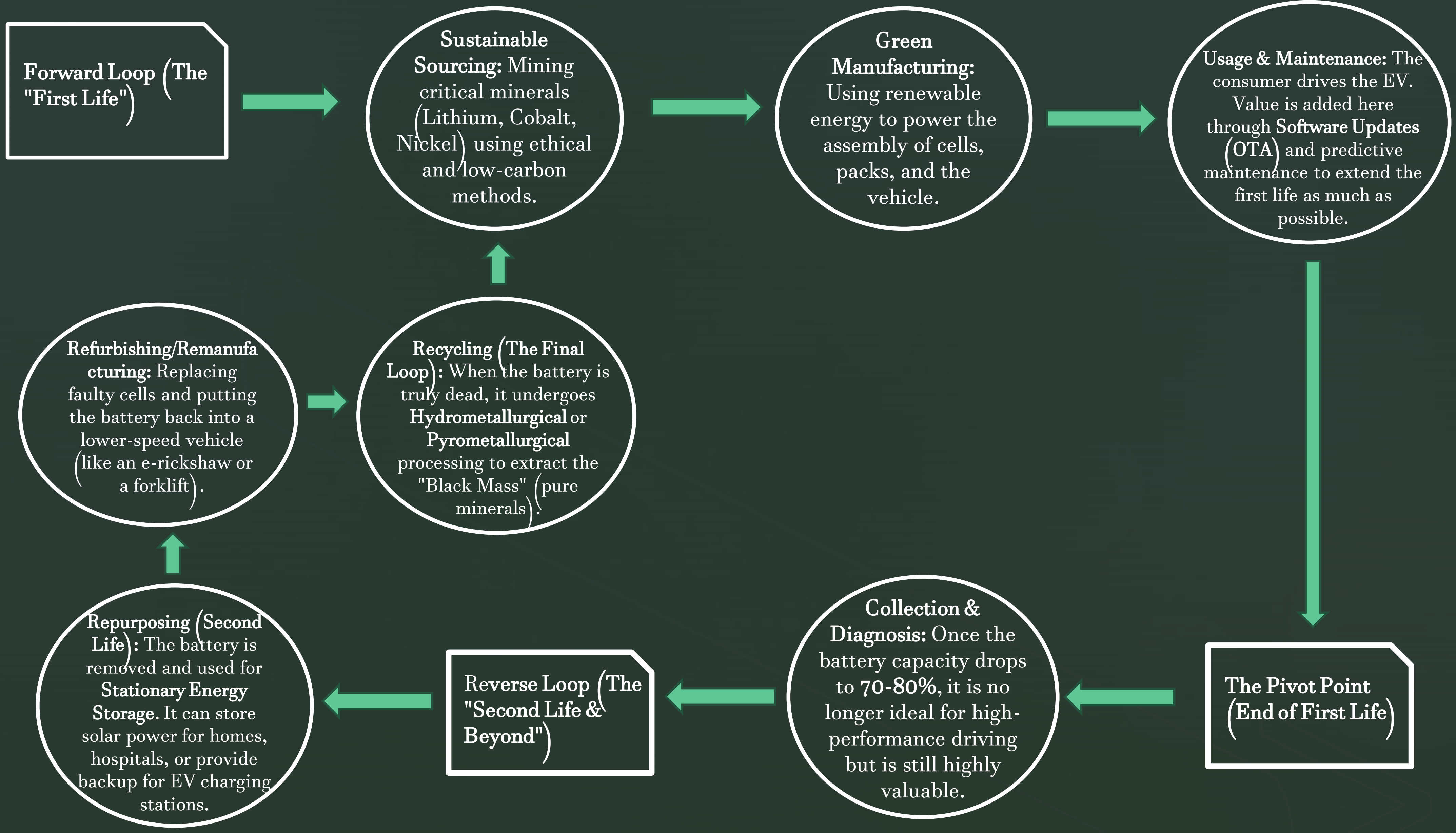
EV Component Cost Breakdown

4. End-of-Life: The Circular Economy

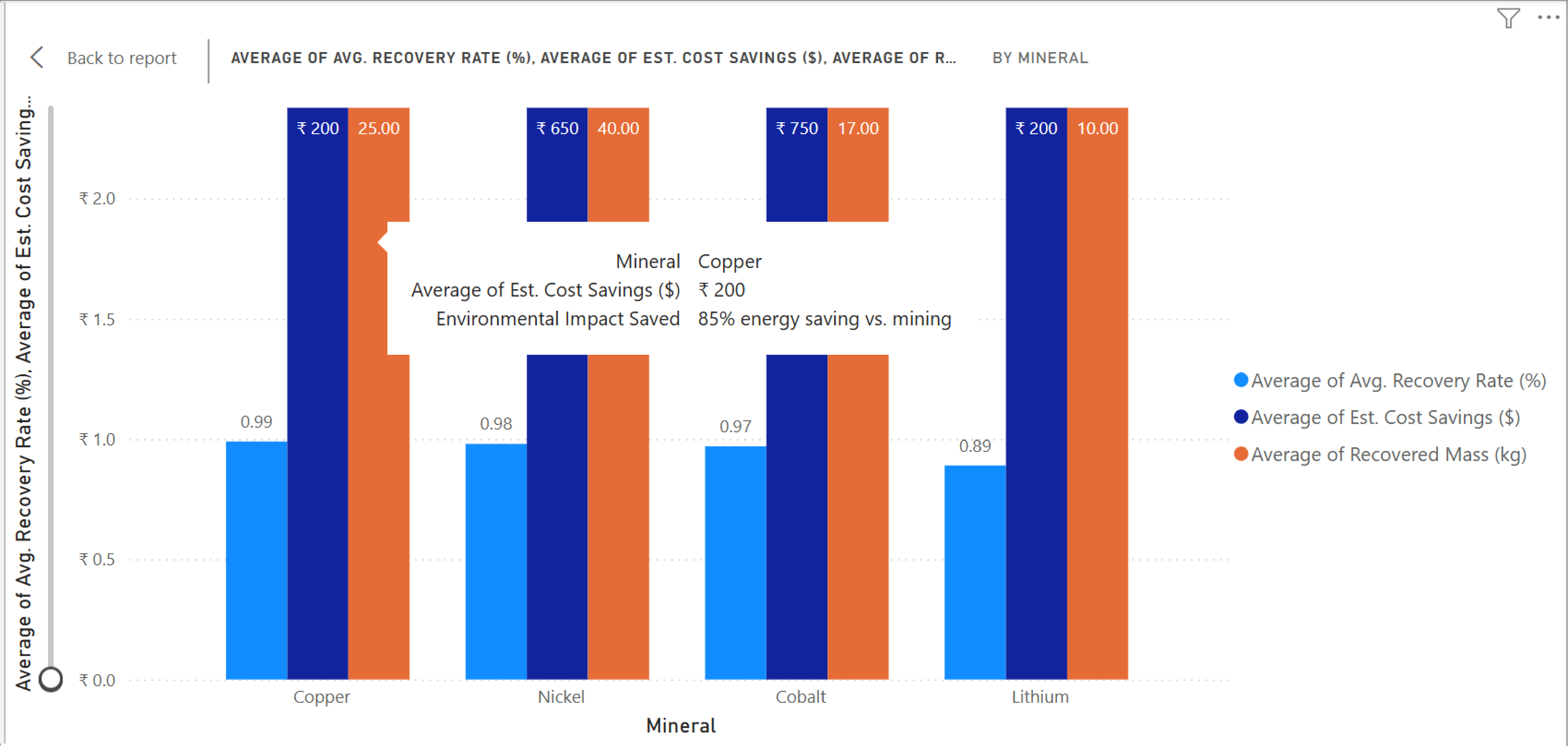
Unlike petrol cars, EVs have a "second life."

- **Second-Life Usage:** Old EV batteries that have dropped to ~70% capacity are repurposed for **Stationary Energy Storage** (e.g., storing solar power for buildings).
- **Recycling (Urban Mining):** Shredding old batteries to recover Lithium, Cobalt, and Nickel. Recovering these is often cheaper and cleaner than mining new ones.

Circular Value Chain



A circular value chain **de-risks the supply chain**. By recycling, an EV company is essentially "mining its own waste," making it immune to global lithium price spikes.



Mineral Recovery & Savings from Circular Value Chain (per 1,000 kg of Battery Waste)

Cost Analysis: TVS iQube (3.1 kWh Variant)

The iQube is marketed for its "ultra-low running costs." For a typical city commute of **30-50 km per day**, here is the data:

- **Charging Cost per KM: ₹ 0.16 to ₹ 0.25** (assuming ₹ 7- ₹ 10 per unit of electricity).
- **Full Charge Cost:** Approximately **₹ 20-₹ 35** depending on the battery variant.
- **Monthly Energy Expense: ~₹ 240- ₹ 315** (for a 1,500 km monthly run).
- **Petrol Comparison:** A 125cc petrol scooter (e.g., TVS Jupiter) costs **~₹ 2.50 per km**.
- **Net Savings:** Switching to iQube saves roughly **₹ 30,000 to ₹ 40,000 per year** in fuel alone.

Total Savings over 50,000 km: ~₹ 93,500(Comparing electricity at ₹ 7/unit vs Petrol at ₹ 100/litre).

1. **Low OpEx (Operating Expenditure):** The running cost is nearly **1/10th** of a petrol scooter.
2. **Mechanical Simplicity:** The lack of a CVT (Continuously Variable Transmission) and Engine Oil systems reduces "Scheduled Maintenance" by **60%**.
3. **Subsidy Dependence:** The **PM E-Drive** reduces the price by roughly **₹15,000–₹20,000**, which is crucial for the 21-month break-even period.

Category	Component	Value (Approx. 2025-26)	Remarks
Fixed Costs	**Ex-Showroom Price**	**₹ 1,08,112**	Includes PM E-Drive Subsidy
	Insurance & Registration	~₹9,500	Varies by State (Delhi/TN are lower)
	On-Road Price (Total)	**₹ 1,17,612**	Initial Capital Expenditure (CAPEX)
Running Costs	Battery Capacity	3.1 kWh	Lithium-ion (NMC)
	Cost per Full Charge	**₹ 21.70**	Assuming ₹7 per unit of electricity
	Real-world Range	100 km	In 'Eco Mode'
	Cost per Kilometer	**₹ 0.22**	vs. ~₹2.50 for Petrol Scooters
Maintenance	Service Interval	4,000 km	Primarily inspections & software
	Annual Service Cost	~₹1,200	Includes consumables like brake oil
	Tyre Replacement	₹ 3,500	Estimated at 25,000 km mark
Long-Term	Battery Warranty	3 Years / 50k km	Standard Manufacturer Warranty
	5-Year Total Savings	**₹ 1,05,000**	Compared to ICE (at 15,000 km/year)
	Break-Even Period	**~21 Months**	Time to recover price gap vs. Petrol

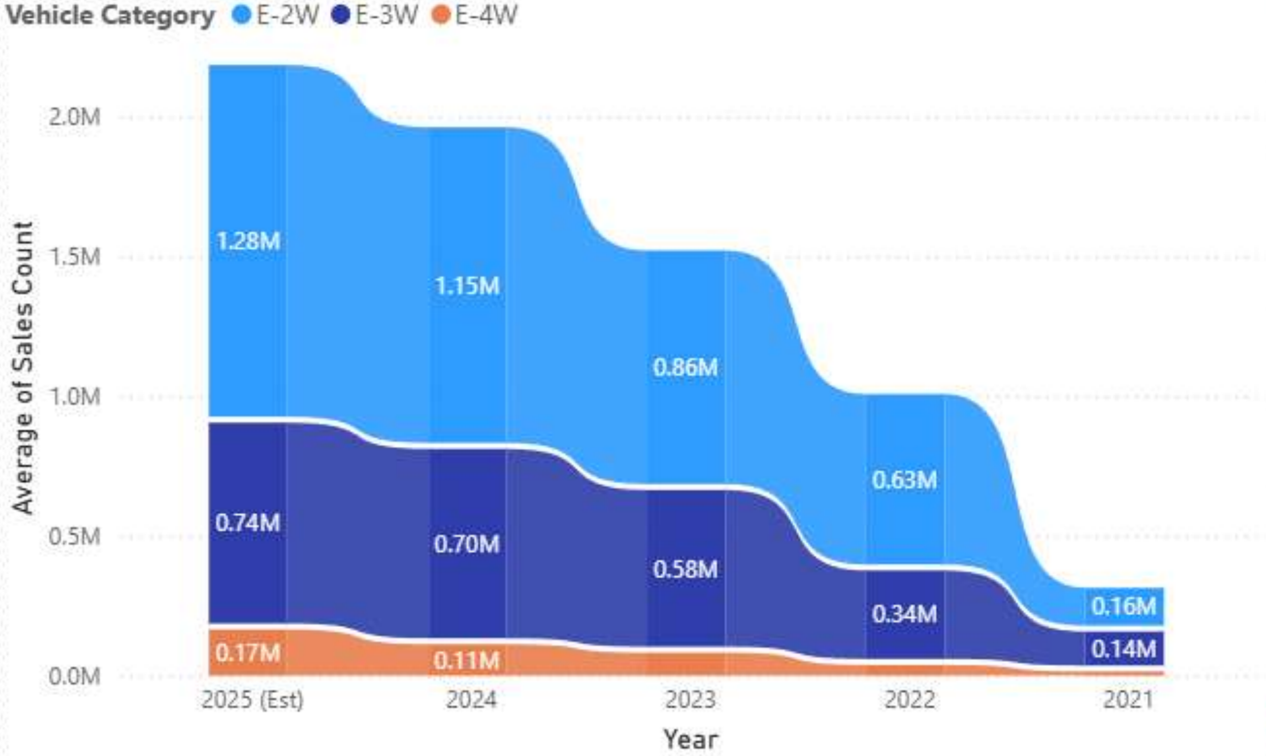
Indian EV Market

3M
Sum of EV_Registration_Count

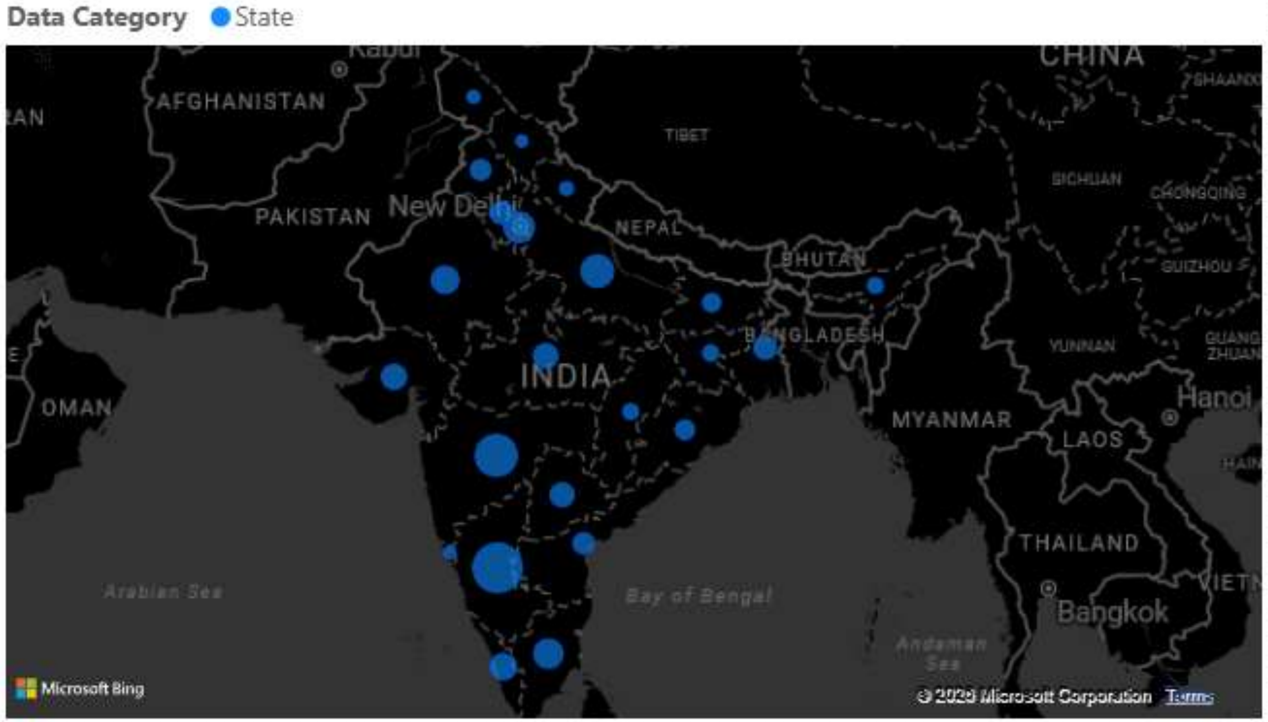
385.00
Average of Avg_Range_km

1335
Sum of Subsidy_Disbursed_Cr

Average of Sales Count by Year and Vehicle Category



Sum of Total EVCS, Sum of Fast Chargers and Sum of Slow Chargers by State/UT and Data Category



Summary of the Indian EV Market Dashboard:

This dashboard quantifies the accelerating EV adoption curve in India, which surpassed 2 million cumulative registrations in late 2025. Key performance indicators (KPIs) reveal an industry-wide Avg. Range of 385 km, marking a technological tipping point for intercity travel. Through a PESTEL-integrated value chain analysis, the data demonstrates that while E-2Wheelers maintain volume dominance (approx. 58%), the E-4Wheeler segment has achieved a 900% growth surge over five years, fueled by the ₹10,900 Cr PM E-DRIVE fiscal stimulus.

Key Takeaways from the Indian EV Market Dashboard:

- The Indian EV market achieved its first '1 Million Year' in 2022 and hit the '2 Million' milestone in late 2025. While E-2Ws account for ~58% of volumes, E-4W growth has surged by over 900% in the last 5 years, signaling the market's shift from niche to mass-market luxury."
- In 2021, the gap between 2-wheelers and 3-wheelers was very small. However, by 2025, the "Ribbon" for E-2W has widened significantly.
- "While 3-wheelers (E-Rickshaws) were the initial drivers of Indian electrification, the market has matured toward personal mobility (E-2W), which now accounts for nearly 60% of total EV registrations."

Challenges and Recommendations:

The transition of India's EV sector into the mass-market phase in 2026 has moved beyond simple consumer awareness. It now faces complex, "second-generation" structural hurdles that require a shift in strategy.

Core Challenges: The "Reliability & Skills Gap“

•**Infrastructure Inconsistency:** A significant "Reliability Gap" exists, with roughly **70% of legacy public charging stations** currently being non-functional or inconsistent.

•**Workforce Readiness:** There is a severe "Skills Gap," as **85% of the traditional automotive workforce** lacks the training required to handle and service high-voltage EV systems.

•**Supply Chain Dependency:** The industry remains heavily reliant on external markets, with an **85% dependency on imported Chinese cells**.

The second-generation challenges and their impact on business

Category	Key Challenges	Impact on Business
Supply Chain	**"The China Dependency"** (85% of cells still imported); Recent **Rare Earth Shocks** causing motor production halts.	High price volatility and risk of production downtime due to geopolitical tensions.
Infrastructure	**"The Reliability Gap"**: Reports show ~70% of legacy Bharat-001 chargers are underutilized or non-functional.	"Charging Anxiety" replaces "Range Anxiety"—users fear finding a charger that actually works.
Financial	**The "Post-Subsidy Cliff"**: As FAME/PM E-DRIVE funds taper off, OEMs struggle to keep prices low.	Shrinking margins for startups; increased pressure on unit economics.
After-Sales	**Skilled Mechanic Shortage**: 85% of traditional mechanics are not trained for high-voltage systems.	Long service turnaround times and poor consumer trust in long-term reliability.
Governance	**Fragmented State Policies**: Differing tax sops across states create a "patchwork" market.	Complex logistics and inventory management for national-level OEMs.

Strategic Recommendations

To succeed in 2026, companies must pivot from "selling a vehicle" to "managing a lifecycle."

A. For Manufacturers (OEMs)

•**Pivot to "Circular Sourcing":** Invest in battery recycling (Urban Mining) to recover Lithium/Cobalt. This reduces dependency on Chinese imports by up to 15%.

•**BaaS (Battery-as-a-Service):** Separate the battery cost from the vehicle price. This slashes upfront costs by 30-40% and solves the "Resale Value" concern by keeping battery ownership with the OEM.

•**Software-Defined Maintenance:** Use the **Data Flywheel** to predict component failure *before* it happens, sending OTA (Over-the-Air) alerts to users to visit service centers.

B. For Charging Point Operators (CPOs)

•**Focus on "Uptime," not just "Number of Plugs":** Shift KPIs from "Total Chargers Installed" to "99% Functional Uptime." Reliability is now a bigger competitive moat than location.

•**Interoperability:** Adopt open standards (like OCPI) so any EV can charge at any station using a single "Super-App."

C. For Policymakers

•**From Incentives to Mandates:** Move away from direct cash subsidies toward **ZLEV (Zero-Low Emission Vehicle) Mandates**, forcing a % of total production to be electric.

•**The "Green Skill" Mission:** Launch national certification programs for EV technicians to build a robust after-sales service ecosystem in Tier-2 and Tier-3 cities.

In 2026, the biggest barrier is no longer **Price** but **Residual Value**. Companies should launch **"Battery Health Passports"**—a digital certificate that proves the battery's health (SOH) to the second-hand buyer. This single feature can increase the resale value of an EV by 15-20%.

EXECUTIVE SUMMARY

•India’s EV industry is entering the **mass-market phase**, driven by policy support, cost parity, and rapid adoption in 2W and 3W segments.

•**E-2Ws dominate volume (~58%)**, while **E-4Ws show explosive growth (900% in 5 years)**, indicating a shift toward higher-value segments.

•**Policy is the strongest adoption driver**, outweighing subsidies; mandates and standards (AIS-156) are reshaping the market.

•**Batteries form 30–40% of EV value**, making supply chains, recycling, and circular economy strategies critical.

•EVs already offer **significantly lower Total Cost of Ownership**, especially in 3W and urban mobility use cases.

•Key challenges are **charging reliability, skill gaps, and import dependence**, requiring lifecycle-focused strategies like recycling, BaaS, and green skilling

Thank You